Production Overview

Total Unit Produced

3,298,062

Average Daily OEE

61.21%

Unplanned Downtime Hours

171.31

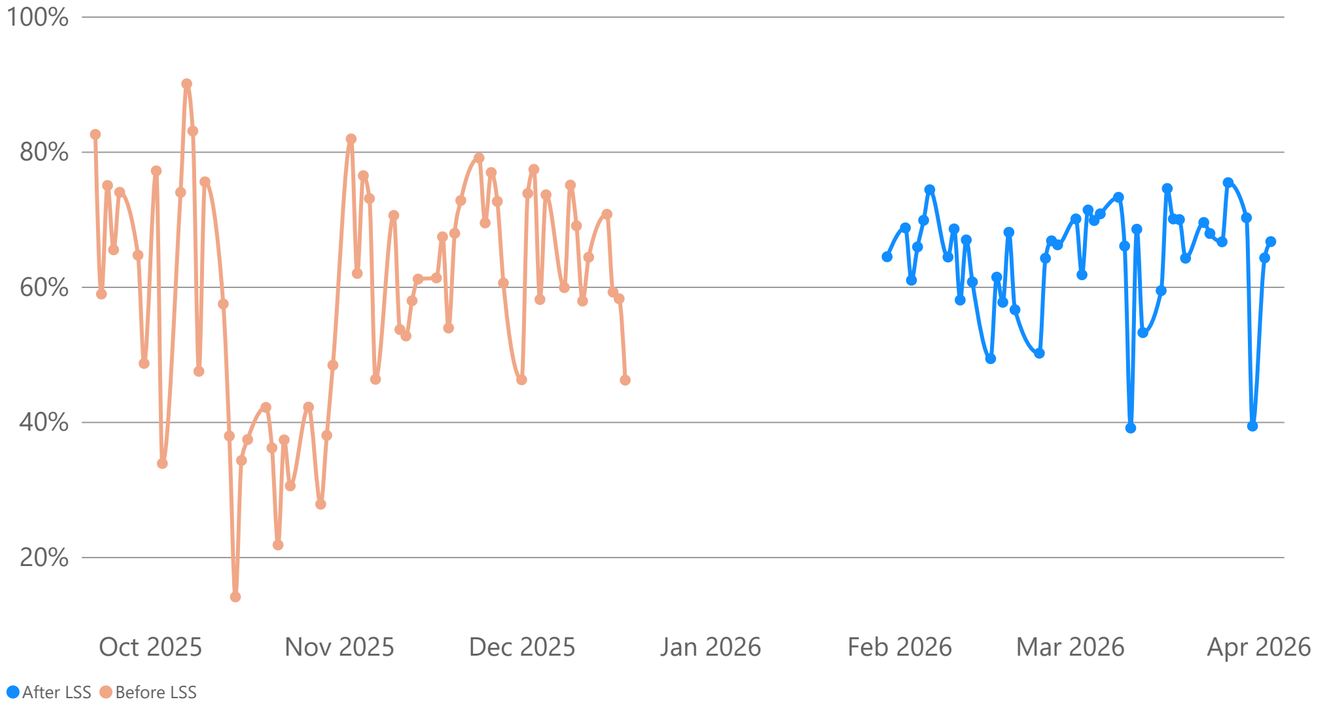
Average Daily Quality

97.82%

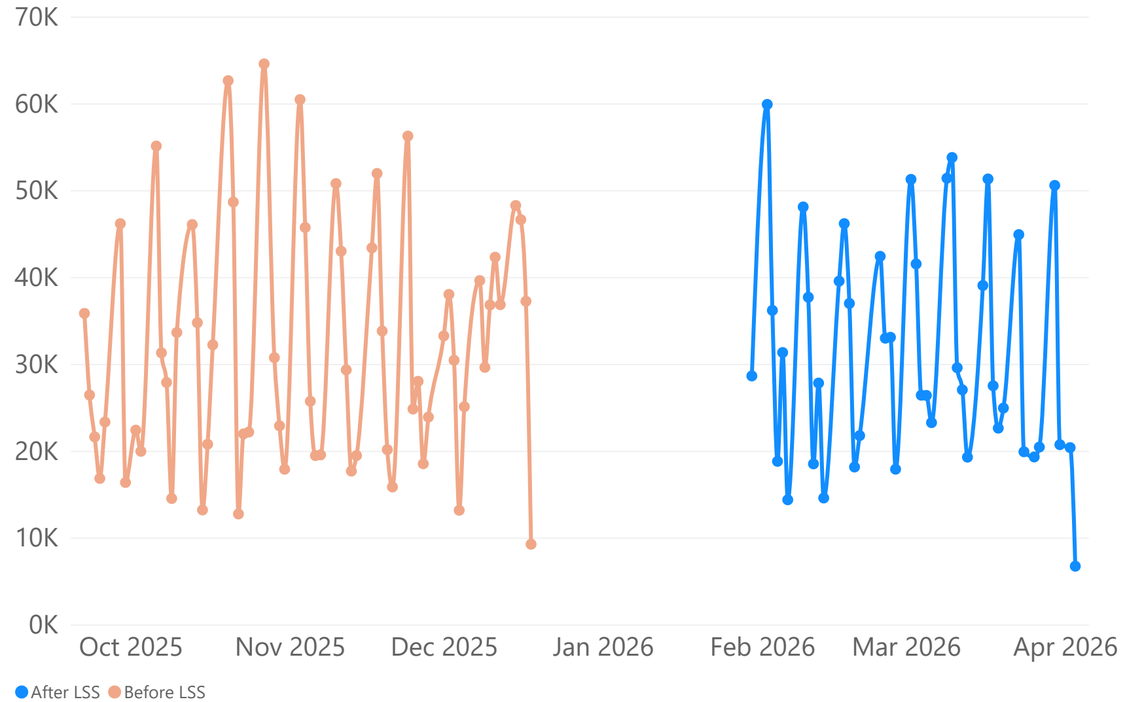
Period Slicer

- Select all
- After LSS
- Before LSS

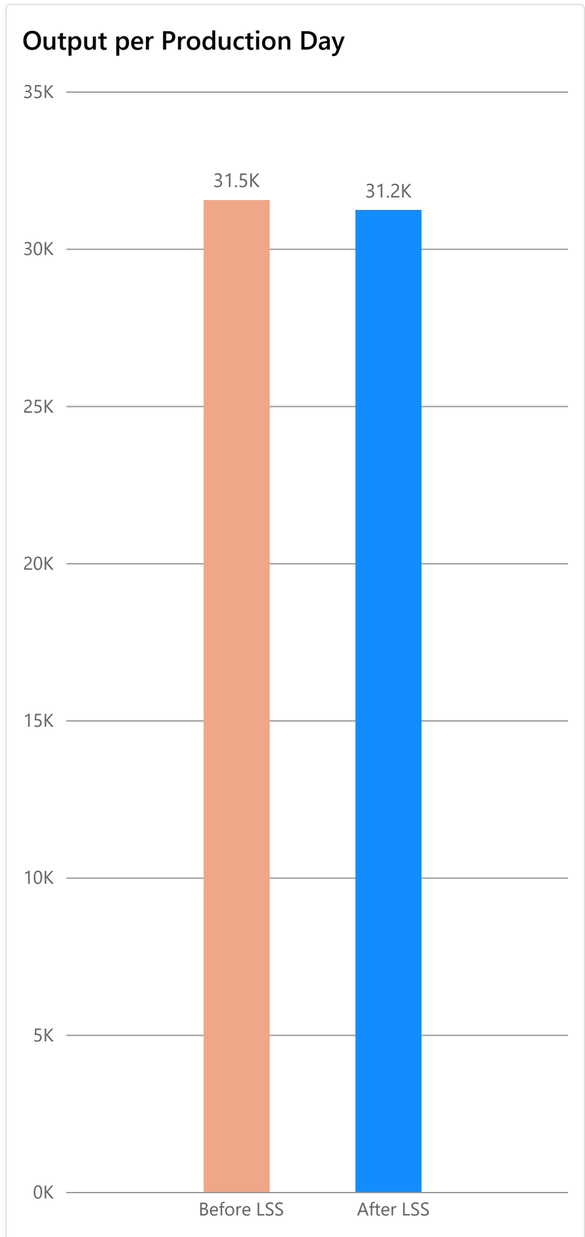
Daily OEE Trend

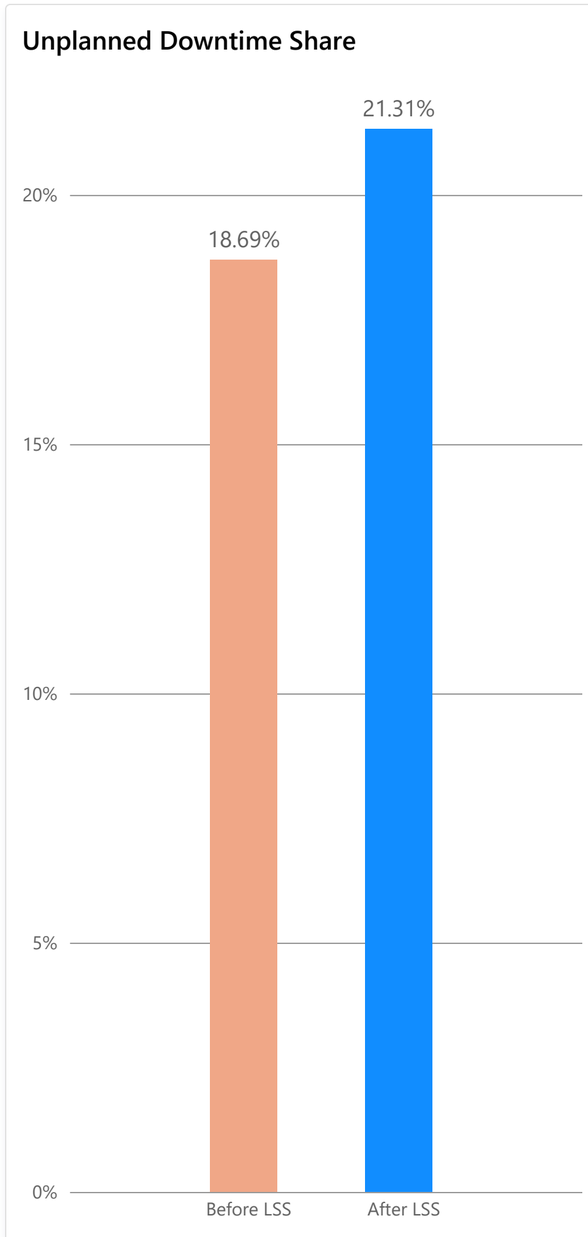


Daily Output



Before And After LSS





Quality Rate

Before LSS	98.50%
After LSS	96.82%

Availability Rate

Before LSS	59.17%
After LSS	69.66%

Performance Rate

Before LSS	93.20%
After LSS	93.74%

Time-Weighted OEE

Before LSS	54.36%
After LSS	63.99%

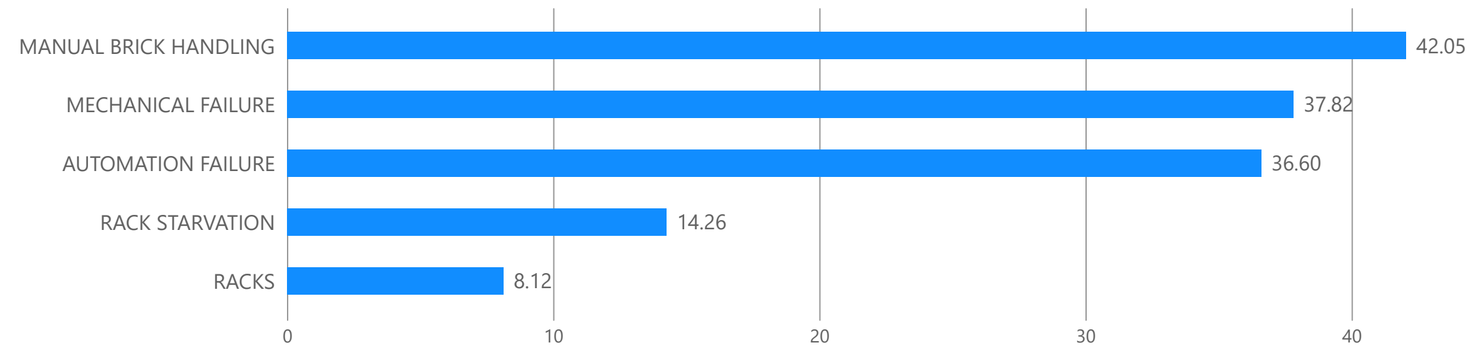
Before LSS: 62 observed production days; After LSS 43.

Downtime Analysis

Period

- Select all
- After LSS
- Before LSS

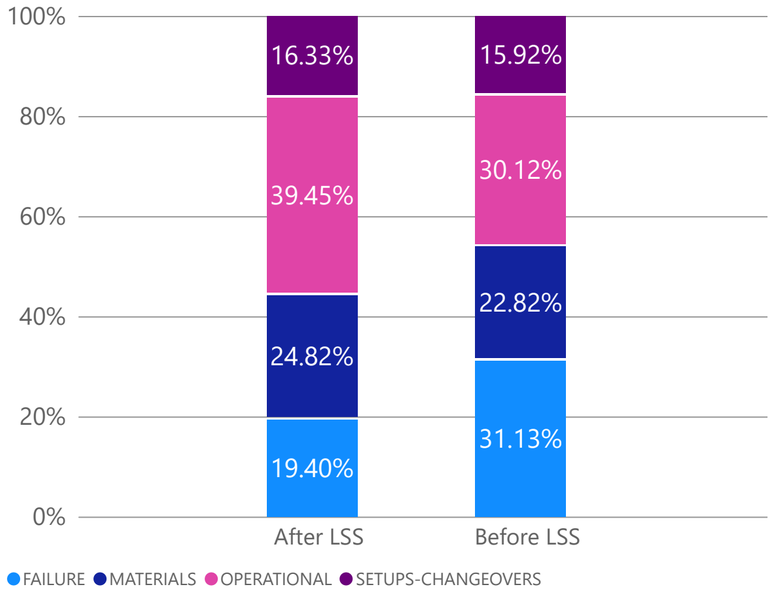
Recorded Stop Hours by Stop



Recorded Stop Hours

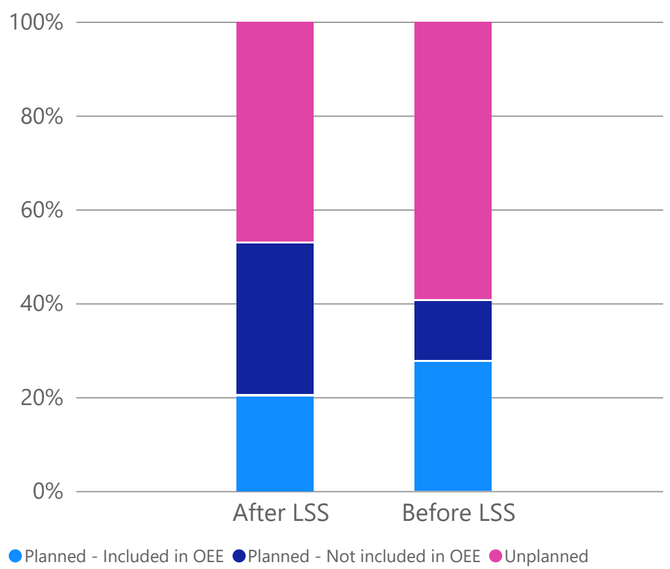
Before LSS	173.13
After LSS	145.17

Stop Group Share by Period and Stop group



Stop Type Time Share

by Period and Stop type



Average Stop Time Duration (min)

Before ...	16.36
After LSS	16.13

Totals cover 62 observed production days before LSS and 43 after LSS.

Sources And Methodology

Source

Independent portfolio project by Caio Bergamo using the OEE and downtime datasets published by Panos Ntoas on Zenodo. The source describes a continuous brick manufacturing line. License: Creative Commons Attribution 4.0 International (CC BY 4.0), as stated in the dataset README. zenodo.org

Dataset: https://doi.org/10.5281/zenodo.17855209

Metric Definitions

- **Time-weighted OEE:** supplied daily OEE weighted by planned production seconds.
- **Availability:** full-speed plus slow-production seconds ÷ planned production seconds.
- **Performance:** total output ÷ ideal production quantity.
- **Quality:** good units ÷ total output.
- **Output per production day:** total output ÷ distinct observed production dates.
- **Unplanned downtime share:** unplanned stop seconds ÷ planned production seconds.
- **Stop group/type time shares:** recorded category duration ÷ total recorded stop duration within each period.
- **Hours:** OEE time fields ÷ 3,600; recorded stop minutes ÷ 60.

Scope & Preparation

Before Lean Six Sigma (LSS): September 22–December 18, 2025; 62 production-date records and 635 stop events.
After LSS: January 30–April 3, 2026; 43 production-date records and 540 stop events.

Prepared in Power Query: removed empty rows and columns, checked data types, standardized column names, added period labels, and appended matching datasets. Greek operator comments were excluded from analysis. Original source files were preserved.

Validation & Limitations

Dates were unique within both OEE files. Calculated event durations were positive; differences between recorded and timestamp-derived durations ranged from −1 to +1 minute across the files. Original durations and unspecified stop-location labels were retained.

Eleven Performance values exceeded 100% and were retained for investigation. Ideal production quantity is provisionally interpreted as units despite the README's seconds label.

The periods have unequal observation lengths. Stop-time shares describe composition, not downtime relative to production time. Before/after differences do not establish that LSS caused the changes. OEE summaries and event logs are analyzed separately; reconciliation and final calculation checks remain pending.